



1. รายละเอียดของโจทย์

1.1 ชื่อโจทย์ (ต้องเป็นภาษาอังกฤษ)

Factory Incident

1.2 หมวดของโจทย์ (Red Team, Blue Team, Miscellaneous)

Miscellaneous

1.3 ชื่อทีม

0xBl4ck-c4t

1.4 ระดับการแข่งขัน (มัธยมศึกษา, อุดมศึกษา)

อุดมศึกษา

1.5 รายชื่อสมาชิกในทีม

หัวหน้าทีม: -

สมาชิกคนที่ 1: -

สมาชิกคนที่ 2: -

2. รายละเอียด Write-up และการแสดงขั้นตอนการแก้ปัญหาจากโจทย์

First I looked at the Protocol Hierarchy first to see what the protocols are in packets.

Wireshark - Protocol Hierarchy Statistics - factory_incident.pcap

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	594	100.0	44474	3,046	0	0	0	594
Ethernet	100.0	594	18.7	8316	569	0	0	0	594
Internet Protocol Version 4	96.0	570	25.6	11400	780	0	0	0	570
User Datagram Protocol	37.9	225	4.0	1800	123	0	0	0	225
Syslog Message	0.3	2	0.6	250	17	2	250	17	2
Simple Network Management Protocol	0.2	1	0.1	43	2	0	0	0	1
Malformed Packet	0.2	1	0.0	0	0	1	0	0	1
Network Time Protocol	1.3	8	0.9	384	26	8	384	26	8
Echo	22.7	135	2.4	1080	73	135	1080	73	135
Dynamic Host Configuration Protocol	0.3	2	1.3	596	40	2	596	40	2
Domain Name System	12.5	74	12.3	5460	374	74	5460	374	74
Data	0.5	3	0.2	91	6	3	91	6	3
Transmission Control Protocol	55.4	329	14.8	6580	450	115	2300	157	329
TPKT - ISO on TCP - RFC1006	0.3	2	0.0	10	0	1	2	0	3
ISO 8073/X.224 COTP Connection-Oriented Transport Protocol	0.3	2	0.3	124	8	0	0	0	2
Remote Desktop Protocol	0.2	1	0.0	8	0	1	8	0	1
Malformed Packet	0.2	1	0.0	0	0	0	0	0	1
Telnet	1.2	7	0.8	345	23	7	345	23	7
Simple Mail Transfer Protocol	2.2	13	2.2	963	65	12	309	21	13
Internet Message Format	0.2	1	1.5	651	44	1	651	44	1
NetBIOS Session Service	0.2	1	0.3	127	8	1	76	5	2
SMB (Server Message Block Protocol)	0.2	1	0.1	47	3	0	0	0	1
Malformed Packet	0.2	1	0.0	0	0	0	0	0	1
MQ Telemetry Transport Protocol	7.2	43	2.7	1222	83	43	1222	83	43
ModbusTCP	21.9	130	4.6	2065	141	0	0	0	130
Modbus	21.9	130	2.6	1155	79	130	1155	79	130
Hypertext Transfer Protocol	1.5	9	2.2	973	66	8	526	36	10
Line-based text data	0.2	1	0.4	200	13	0	0	0	1
HTML Form URL Encoded	0.2	1	0.1	32	2	1	32	2	1
File Transfer Protocol (FTP)	1.5	9	1.3	577	39	9	577	39	9
Internet Control Message Protocol	2.7	16	2.4	1072	73	8	320	21	16
HiPerConTracer Trace Service	1.3	8	1.5	688	47	8	688	47	8
Address Resolution Protocol	4.0	24	1.5	672	46	24	672	46	24

No display filter.

Close Copy Protocols Help

We can see that mostly it was UDP and TCP.

So I filtered out the UDP first to see what the packets would be.

factory_incident.pcap

No.	Time	Source	S.Port	Destination	T.Port	Protocol	Length	Host	Identification	Info
14	2025-09-05 15:33:22.795002	10.10.1.1	67	10.10.1.50	67	DHCP	230		0x0001	DHCP Offer - Transaction ID 0x9056cc71
15	2025-09-05 15:33:22.993756	10.10.1.50	57591	10.10.1.2	53	DNS	81		0x0001	Standard query 0x0000 A fileservr.corp.local
16	2025-09-05 15:33:23.047110	10.10.1.2	53	10.10.1.50	57591	DNS	110		0x0001	Standard query response 0x0000 A fileservr.corp.local A 10.10.1.10
17	2025-09-05 15:33:23.120332	10.10.1.50	50703	10.10.1.2	53	DNS	75		0x0001	Standard query 0x0000 A mail.corp.local
18	2025-09-05 15:33:23.468983	10.10.1.2	53	10.10.1.50	50703	DNS	196		0x0001	Standard query response 0x0000 A mail.corp.local A 10.10.1.10
19	2025-09-05 15:33:23.555372	10.10.1.50	42217	10.10.1.2	53	DNS	80		0x0001	Standard query 0x0000 A printer01.corp.local
20	2025-09-05 15:33:23.634062	10.10.1.2	53	10.10.1.50	42217	DNS	116		0x0001	Standard query response 0x0000 A printer01.corp.local A 10.10.1.10
21	2025-09-05 15:33:23.746455	10.10.1.50	43481	10.10.1.2	53	DNS	88		0x0001	Standard query 0x0000 TXT maintenance-notes.corp.local
22	2025-09-05 15:33:23.888778	10.10.1.2	53	10.10.1.50	43481	DNS	206		0x0001	Standard query response 0x0000 TXT maintenance-notes.corp.local TXT [Malformed Packet]
22a	2025-09-05 15:34:03.706050	10.10.5.21	55123	10.10.5.2	161	SNMP	164		0x0001	
22b	2025-09-05 15:34:03.921252	10.10.5.2	514	10.10.5.21	514	Syslog	176		0x0001	LOCAL.INFO: Sep 11 08:01:00 CoreSwitch-01: link-up Gi0/3 note:CYBERHEI
260	2025-09-05 15:34:09.394391	10.10.5.99	53711	10.10.5.100.77	53	DNS	113		0x0001	Standard query 0x0000 A CYBERHEROCTF-secure-relay-decoy-flag-not-real.c
261	2025-09-05 15:34:09.498890	10.10.5.100.77	53	10.10.5.99	53711	DNS	182		0x0001	Standard query response 0x0000 A CYBERHEROCTF-secure-relay-decoy-flag-not-real.c
262	2025-09-05 15:34:09.625912	10.10.5.2	514	10.10.5.21	514	Syslog	158		0x0001	LOCAL.INFO: Sep 11 08:07:00 IDS-sensor: sig=decoy-flag-domain on 10.10.5.21
271	2025-09-05 15:34:11.773828	10.10.5.99	31923	10.10.5.21	7	ECHO	50		0x0431	Request
272	2025-09-05 15:34:11.839873	10.10.5.99	34968	10.10.5.21	7	ECHO	50		0x102a	Request
273	2025-09-05 15:34:11.916902	10.10.5.99	31968	10.10.5.21	7	ECHO	50		0x0637	Request
274	2025-09-05 15:34:12.237682	10.10.5.99	32484	10.10.5.21	7	ECHO	50		0x1b02	Request
275	2025-09-05 15:34:12.331822	10.10.5.99	38439	10.10.5.21	7	ECHO	50		0x0c18	Request
276	2025-09-05 15:34:12.598052	10.10.5.99	39342	10.10.5.21	7	ECHO	50		0x176c	Request
277	2025-09-05 15:34:12.634222	10.10.5.99	36389	10.10.5.21	7	ECHO	50		0x106f	Request
278	2025-09-05 15:34:12.774717	10.10.5.99	32564	10.10.5.21	7	ECHO	50		0x1f55	Request
279	2025-09-05 15:34:12.986162	10.10.5.99	37768	10.10.5.21	7	ECHO	50		0x2b4a	Request
280	2025-09-05 15:34:13.223385	10.10.5.99	35433	10.10.5.21	7	ECHO	50		0x2a31	Request
281	2025-09-05 15:34:13.264235	10.10.5.99	35184	10.10.5.21	7	ECHO	50		0x236c	Request
282	2025-09-05 15:34:13.533832	10.10.5.99	37119	10.10.5.21	7	ECHO	50		0x0a97	Request
283	2025-09-05 15:34:13.639562	10.10.5.99	32247	10.10.5.21	7	ECHO	50		0x1c0f	Request
284	2025-09-05 15:34:13.745893	10.10.5.99	39832	10.10.5.21	7	ECHO	50		0x0d11	Request
285	2025-09-05 15:34:13.981912	10.10.5.99	32035	10.10.5.21	7	ECHO	50		0x1226	Request
286	2025-09-05 15:34:14.138902	10.10.5.99	35682	10.10.5.21	7	ECHO	50		0x087f	Request
287	2025-09-05 15:34:14.437631	10.10.5.99	31184	10.10.5.21	7	ECHO	50		0x0973	Request
288	2025-09-05 15:34:14.722250	10.10.5.99	33657	10.10.5.21	7	ECHO	50		0x2044	Request
289	2025-09-05 15:34:15.062260	10.10.5.99	34209	10.10.5.21	7	ECHO	50		0x0376	Request
290	2025-09-05 15:34:15.061644	10.10.5.99	37625	10.10.5.21	7	ECHO	50		0x161e	Request
291	2025-09-05 15:34:15.288073	10.10.5.99	36652	10.10.5.21	7	ECHO	50		0x2459	Request
292	2025-09-05 15:34:15.636157	10.10.5.99	33535	10.10.5.21	7	ECHO	50		0x2606	Request
293	2025-09-05 15:34:15.720950	10.10.5.99	31064	10.10.5.21	7	ECHO	50		0x2e47	Request
294	2025-09-05 15:34:15.761812	10.10.5.99	38867	10.10.5.21	7	ECHO	50		0x0917	Request
295	2025-09-05 15:34:15.998972	10.10.5.99	31232	10.10.5.21	7	ECHO	50		0x152c	Request
296	2025-09-05 15:34:16.228349	10.10.5.99	34721	10.10.5.21	7	ECHO	50		0x0211	Request
297	2025-09-05 15:34:16.410712	10.10.5.99	37280	10.10.5.21	7	ECHO	50		0x1129	Request
298	2025-09-05 15:34:16.486624	10.10.5.99	38960	10.10.5.21	7	ECHO	50		0x135d	Request
299	2025-09-05 15:34:16.727618	10.10.5.99	33084	10.10.5.21	7	ECHO	50		0x2840	Request
300	2025-09-05 15:34:17.061592	10.10.5.99	37196	10.10.5.21	7	ECHO	50		0x1546	Request
301	2025-09-05 15:34:17.102622	10.10.5.99	32115	10.10.5.21	7	ECHO	50		0x1a35	Request
302	2025-09-05 15:34:17.360262	10.10.5.99	39452	10.10.5.21	7	ECHO	50		0x2540	Request
303	2025-09-05 15:34:17.583982	10.10.5.99	31607	10.10.5.21	7	ECHO	50		0x2c13	Request
304	2025-09-05 15:34:17.634052	10.10.5.99	39104	10.10.5.21	7	ECHO	50		0x1e46	Request
305	2025-09-05 15:34:17.934823	10.10.5.99	36722	10.10.5.21	7	ECHO	50		0x2221	Request
306	2025-09-05 15:34:18.269151	10.10.5.99	32627	10.10.5.21	7	ECHO	50		0x010d	Request

We can see here there's something suspicious with the ECHO protocol that is going on a lot. So I decided to narrow it down so it's easier to see.

Filter: ip.src == 10.10.5.99 && ip.dst == 10.10.5.21 &&
udp.dstport == 7

No.	Time	Source	S_Port	Destination	T_Port	Protocol	Length	Host	Identification	Info
274	2025-09-05 15:34:11.7782411	10.10.5.99	31823	10.10.5.21	7	ECHO	50		0x0421	Request
272	2025-09-05 15:34:11.8398732	10.10.5.99	34968	10.10.5.21	7	ECHO	50		0x1d2a	Request
273	2025-09-05 15:34:11.9169992	10.10.5.99	31869	10.10.5.21	7	ECHO	50		0x0637	Request
274	2025-09-05 15:34:12.2376822	10.10.5.99	32484	10.10.5.21	7	ECHO	50		0x1b62	Request
275	2025-09-05 15:34:12.3318222	10.10.5.99	38439	10.10.5.21	7	ECHO	50		0x8c18	Request
276	2025-09-05 15:34:12.5988552	10.10.5.99	39342	10.10.5.21	7	ECHO	50		0x176c	Request
277	2025-09-05 15:34:12.6342222	10.10.5.99	36389	10.10.5.21	7	ECHO	50		0x106f	Request
278	2025-09-05 15:34:12.7747172	10.10.5.99	32564	10.10.5.21	7	ECHO	50		0x1f55	Request
279	2025-09-05 15:34:12.9861662	10.10.5.99	37788	10.10.5.21	7	ECHO	50		0x2b4a	Request
280	2025-09-05 15:34:13.2233852	10.10.5.99	35433	10.10.5.21	7	ECHO	50		0x2a31	Request
281	2025-09-05 15:34:13.2642352	10.10.5.99	35184	10.10.5.21	7	ECHO	50		0x236c	Request
282	2025-09-05 15:34:13.5338232	10.10.5.99	37119	10.10.5.21	7	ECHO	50		0x0a87	Request
283	2025-09-05 15:34:13.6395662	10.10.5.99	32247	10.10.5.21	7	ECHO	50		0x1c8f	Request
284	2025-09-05 15:34:13.7458932	10.10.5.99	39832	10.10.5.21	7	ECHO	50		0x0d11	Request
285	2025-09-05 15:34:13.8910912	10.10.5.99	32835	10.10.5.21	7	ECHO	50		0x1226	Request
286	2025-09-05 15:34:14.1380982	10.10.5.99	35682	10.10.5.21	7	ECHO	50		0x087f	Request
287	2025-09-05 15:34:14.4378312	10.10.5.99	31184	10.10.5.21	7	ECHO	50		0x0073	Request
288	2025-09-05 15:34:14.7222562	10.10.5.99	33697	10.10.5.21	7	ECHO	50		0x2844	Request
289	2025-09-05 15:34:15.0822682	10.10.5.99	34289	10.10.5.21	7	ECHO	50		0x0376	Request
290	2025-09-05 15:34:15.0816442	10.10.5.99	37625	10.10.5.21	7	ECHO	50		0x161e	Request
291	2025-09-05 15:34:15.2886732	10.10.5.99	36652	10.10.5.21	7	ECHO	50		0x2450	Request
292	2025-09-05 15:34:15.6361572	10.10.5.99	33535	10.10.5.21	7	ECHO	50		0x2686	Request
293	2025-09-05 15:34:15.7289592	10.10.5.99	31864	10.10.5.21	7	ECHO	50		0x2e47	Request
294	2025-09-05 15:34:15.7618122	10.10.5.99	38887	10.10.5.21	7	ECHO	50		0x0917	Request
295	2025-09-05 15:34:15.9998072	10.10.5.99	31282	10.10.5.21	7	ECHO	50		0x192c	Request
296	2025-09-05 15:34:16.2283492	10.10.5.99	34721	10.10.5.21	7	ECHO	50		0x0211	Request
297	2025-09-05 15:34:16.4187712	10.10.5.99	37288	10.10.5.21	7	ECHO	50		0x1129	Request
298	2025-09-05 15:34:16.4968642	10.10.5.99	38888	10.10.5.21	7	ECHO	50		0x135a	Request
299	2025-09-05 15:34:16.7278182	10.10.5.99	33884	10.10.5.21	7	ECHO	50		0x2848	Request
300	2025-09-05 15:34:17.0615922	10.10.5.99	37196	10.10.5.21	7	ECHO	50		0x1546	Request
301	2025-09-05 15:34:17.1625422	10.10.5.99	32115	10.10.5.21	7	ECHO	50		0x1a35	Request
302	2025-09-05 15:34:17.3682862	10.10.5.99	39452	10.10.5.21	7	ECHO	50		0x2548	Request
303	2025-09-05 15:34:17.5838882	10.10.5.99	31687	10.10.5.21	7	ECHO	50		0x2c13	Request
304	2025-09-05 15:34:17.6348552	10.10.5.99	39184	10.10.5.21	7	ECHO	50		0x1e46	Request
305	2025-09-05 15:34:17.9348232	10.10.5.99	36722	10.10.5.21	7	ECHO	50		0x2221	Request
306	2025-09-05 15:34:18.2681512	10.10.5.99	32527	10.10.5.21	7	ECHO	50		0x018d	Request
307	2025-09-05 15:34:18.5884882	10.10.5.99	39889	10.10.5.21	7	ECHO	50		0x148d	Request
308	2025-09-05 15:34:18.7339282	10.10.5.99	36875	10.10.5.21	7	ECHO	50		0x2167	Request
309	2025-09-05 15:34:18.8783142	10.10.5.99	37597	10.10.5.21	7	ECHO	50		0x298b	Request
310	2025-09-05 15:34:18.9562982	10.10.5.99	32424	10.10.5.21	7	ECHO	50		0x053d	Request
311	2025-09-05 15:34:19.2189952	10.10.5.99	36225	10.10.5.21	7	ECHO	50		0x384d	Request
312	2025-09-05 15:34:19.5448522	10.10.5.99	38256	10.10.5.21	7	ECHO	50		0x0f86	Request
313	2025-09-05 15:34:19.6348422	10.10.5.99	31188	10.10.5.21	7	ECHO	50		0x2783	Request
314	2025-09-05 15:34:19.7882982	10.10.5.99	38291	10.10.5.21	7	ECHO	50		0x1883	Request
315	2025-09-05 15:34:19.9589152	10.10.5.99	31156	10.10.5.21	7	ECHO	50		0x0761	Request
316	2025-09-05 15:34:20.0232382	10.10.5.99	31229	10.10.5.21	7	ECHO	50		0x2786	Request
317	2025-09-05 15:34:20.3815882	10.10.5.99	31789	10.10.5.21	7	ECHO	50		0x0b75	Request
318	2025-09-05 15:34:20.3745262	10.10.5.99	31971	10.10.5.21	7	ECHO	50		0x2d41	Request
319	2025-09-05 15:34:20.4978842	10.10.5.99	33895	10.10.5.21	7	ECHO	50		0x0e1c	Request

I inspected the packet contents and noticed that the UDP
payload contained values that could be used as sequence
numbers

No.	Time	Source	S_Port	Destination	T_Port	Protocol	Length	Host	Identification	Info
271	2025-09-05 15:34:11.7788262	10.10.5.99	31823	10.10.5.21	7	ECHO	50		0x0421	Request
272	2025-09-05 15:34:11.8338732	10.10.5.99	34968	10.10.5.21	7	ECHO	50		0x1d2a	Request
273	2025-09-05 15:34:11.9169992	10.10.5.99	31869	10.10.5.21	7	ECHO	50		0x0637	Request
274	2025-09-05 15:34:12.2376822	10.10.5.99	32484	10.10.5.21	7	ECHO	50		0x1b62	Request
Frame 271: Packet, 50 bytes on wire (400 bits), 50 bytes captured (400 bits) on interface 0										
Ethernet II, Src: 02:00:00:00:00:aa (02:00:00:00:00:aa), Dst: 02:00:00:00:00:bb (02:00:00:00:00:bb)										
Internet Protocol Version 4, Src: 10.10.5.99, Dst: 10.10.5.21										
0100 5 Version: 4										
.... 0101 = Header Length: 20 bytes (5)										
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)										
Total Length: 36										
Identification: 0x0431 (1073)										
0000 02 00 00 00 00 bb 02 00 00 00 00 aa 08 00 45 00										
0010 00 24 04 31 00 00 40 11 58 6d 0a 0a 05 63 0a 0a										
0030 05 1c b3 00 07 00 10 68 65 64 00 00 00 00 00										
0030 00 00										

I use the ip.id as the filter columns

Identification
0x0431
0x1d2a
0x0637
0x1b02
0x0c18
0x176c
0x106f
0x1f55
0x2b4a
0x2a31
0x236c
0x0a07
0x1c0f
0x0d11
0x1226
0x087f
0x0073
0x2044
0x0376
0x161e
0x2450
0x2606
0x2e47
0x0917
0x192c
0x0211
0x1120
0x135d
0x2840
0x1546
0x1a35
0x2540
0x2c13
0x1e46
0x2221
0x010d
0x140d
0x2167
0x290b
0x053d
0x304d
0x0f06
0x2703
0x1803
0x0761
0x2f06
0x0b75
0x2d41
0x0e1c

I wrote a python code to automate the extraction of the hidden data in the packets and decode out the flag:

```
import subprocess
```

```
# command to get the ip.id to calculate the plaintext
```

```
cmd = [  
    "tshark",  
    "-r", "factory_incident.pcap",  
    "-Y", "ip.src==10.10.5.99 && ip.dst==10.10.5.21 &&  
    udp.dstport==7",  
    "-T", "fields",  
    "-e", "frame.number",  
    "-e", "ip.id",  
    "-e", "udp.payload"  
]
```

```
output = subprocess.check_output(cmd, text=True)
```

```
packets = []
```

```
for line in output.splitlines():  
    packet, ip_id, payload = line.split("\t")
```

```
'''
```

The first byte of each UDP payload acts as a sequence number.

payload[:2] takes the first two hexadecimal characters, and int(..., 16) converts them from hexadecimal to an integer.

```
'''
```

```
    # First byte of UDP payload = sequence number  
    sequence = int(payload[:2], 16)
```

```

# Low byte of IP ID = hidden data
'''
The IPv4 Identification field contains the hidden data. I
converted the IP ID from hexadecimal to an integer and used &
0xff to keep only its lowest 8 bits.
'''

hidden_byte = int(ip_id, 16) & 0xff

packets.append((sequence, int(packet), hidden_byte))

#sorted them using the sequence number extracted from the
UDP payload packets.sort()

// print out the
for sequence, packet, hidden_byte in packets:
    print(f'{sequence:02x} packet={packet}
data={hidden_byte:02x}')

# I grabbed the hidden bytes from each packets then assemble
it into one ciphertext
ciphertext = bytes(x[2] for x in packets)

print("\nCiphertext:")
print(ciphertext.hex())

ciphertext = bytes.fromhex(ciphertext.hex())

#the flag format is known, I used CYBERHEROCTF{ as known
plaintext
known = b"CYBERHEROCTF{"

```

```
# XORing the known plaintext against the corresponding  
ciphertext bytes reveals the beginning of the XOR key
```

```
key = bytes(  
    ciphertext[i] ^ known[i]  
    for i in range(len(known))  
)
```

```
print(key)
```

```
#We take just the first 8 bytes because it's a repeated key  
key = key[:8]
```

```
#XOR the key with the ciphertext
```

```
plaintext = bytes(  
    ciphertext[i] ^ key[i % len(key)]  
    for i in range(len(ciphertext))  
)
```

```
#print out the flag
```

```
print(plaintext.decode())
```

The cipher text I got:

730d1176313d37617f17077518111c066f20265d0d461e6c032c
35020f2a46554467216c50400603400b314a134147064d

Repeated key: 0TS3cur30TS3c

Key that use to XOR with the cipher text: 0TS3cur3

Flag:

CYBERHEROCTF{dn5_tunn3l_3xf1l_4ft3r_35t0p_byp455}

```
python solve.py - OxBi4ck-c4t
00 packet=287 data=73
01 packet=306 data=0d
02 packet=296 data=11
03 packet=289 data=76
04 packet=271 data=31
05 packet=310 data=3d
06 packet=273 data=37
07 packet=315 data=61
08 packet=286 data=7f
09 packet=294 data=17
0a packet=282 data=07
0b packet=317 data=75
0c packet=275 data=18
0d packet=284 data=11
0e packet=319 data=1c
0f packet=312 data=06
10 packet=277 data=6f
11 packet=297 data=20
12 packet=285 data=26
13 packet=298 data=5d
14 packet=307 data=0d
15 packet=300 data=46
16 packet=290 data=1e
17 packet=276 data=6c
18 packet=314 data=03
19 packet=295 data=2c
1a packet=301 data=35
1b packet=274 data=02
1c packet=283 data=0f
1d packet=272 data=2a
1e packet=304 data=46
1f packet=278 data=55
20 packet=288 data=44
21 packet=308 data=67
22 packet=305 data=21
23 packet=281 data=6c
24 packet=291 data=50
25 packet=302 data=40
26 packet=292 data=06
27 packet=313 data=03
28 packet=299 data=40
29 packet=309 data=0b
2a packet=280 data=31
2b packet=279 data=4a
2c packet=303 data=13
2d packet=318 data=41
2e packet=293 data=47
2f packet=316 data=06
30 packet=311 data=4d

Ciphertext:
730d1176313d37617f17077518111c066f20265d0d461e6c032c35020f2a46554467216c50400603400b314a134147064d

packets.append((sequence, int(
hidden_byte))

packets.sort()

// print out the
for sequence, packet, hidden_byte
    print(f'{sequence:02x} packet
data={hidden_byte:02x}')

ciphertext = bytes(x[2] for x in pa

print("\nCiphertext:")
print(ciphertext.hex())

ciphertext = bytes.fromhex(cipher

known = b"CYBERHEROCTF{"

key = bytes(
    ciphertext[i] ^ known[i]
    for i in range(len(known))
)
```

```
Ciphertext:
730d1176313d37617f17077518111c066f20265d0d461e6c032c35020f2a46554467216c50400603400b314a134147064d
b'0TS3cur30TS3c'
CYBERHEROCTF{dn5_tunn3l_3xf1l_4ft3r_35t0p_byp455}

known = b"CYBERHEROCTF{"
```